RAYMOND M. VALDEPEÑAS

BS in Computer Applications Major in IOT

Contact: 09667324567 – mond.valdepenas.m.13@gmail.com
Address: Purok 3, Sta. Cruz, Tangub City, Misamis Occidental, Philippines



About Me

Full-Stack Developer with experience in building and deploying SaaS, IoT-enabled, and scalable applications using React/Next.js, Node.js, and MySQL. Skilled in integrating AI and IoT technologies, with research published under Springer NiDS 2025. Adept at managing full-cycle development from concept to deployment. Passionate about leveraging technology to solve real-world problems and continuously learning new frameworks and tools.

Tech Stack

Frontend:
- HTML5/CSS3
- TypeScript/JavaScript
- React.js/Next.js
- Zustand
- Ant Design
- ShadCN
- Material UI
- TailwindCSS

Backend:
- Node.js/Express.js
- MySQL/PostgreSQL
- Flask
- FastAPI
- Redis
- RESTful APIs
- WebSockets
- NextAuth.js

Tools:
- VS Code
- Claude AI
- Git/Github
- CI/CD
- NodeMailer
- Firebase
- Vercel/Render
- ESLint

Work Experience

Internship Logicbase Interactive Ent., Cagayan de Oro City, Philippines (Jan 2025 - May 2025)

- **Developed and deployed** a Next.js web app for project management and customer support, to enhance team efficiency and collaboration.
- **Conducted comprehensive QA testing** on a POS application, identifying and documenting 15+ critical and 100+ minor bugs, which accelerated issue resolution and improved overall system stability and user experience.
- **Designed and prototyped** a web app landing page using Figma.
- **Assisted** in creating digital marketing content that increased company visibility across 3+ social media platforms.

Education

College Mindanao State University - Iligan Institute of Technology (2021-2025)
Bachelor of Science in Computer Applications Major in Internet of Things
Grade: **1.42535**
Honor: **Magna Cum Laude**

Senior High School Northwestern Mindanao State College of Science and Technology (2019-2021)

Junior High School Lorenzo Tan National High School (2015-2019)

Projects

Logicbase Command – Internal Project Management and Customer Support Web App
Designed and developed a Next.js + MySQL web application for internal project tracking and client support, improving task visibility and team coordination by 25% and reduced client response times by 30% through streamlined issue tracking and role-based access.

MoneyCacheHub – Personal Finance Management Web App
Built using Next.js, Zustand, MySQL, and NextAuth.js for authentication. Implemented real-time transaction tracking and PDF generation with @react-pdf/renderer. Deployed on Vercel, achieving consistent uptime and smooth user experience with global state sync.

POS Transactions Management System – Logicbase OJT Project
Developed and deployed a Next.js + MySQL POS management platform with modular backend architecture. Enabled transaction tracking, reporting, and secure authentication, deployed on Vercel for client use.

Mobile Autonomous Plant Watering AI-Robot
A robotics project designed to automate plant care using an AI-Powered Robot equipped with a Raspberry Pi 5, Arducam Camera, various sensors, and other components. The robot can navigate autonomously, avoid obstacles, and water plants using computer vision. It utilizes YOLOv11 (with over 90% accuracy) to detect, classify, and track plants, ensuring accurate watering. A web app provides a control dashboard for scheduling and monitoring watering tasks.

my.BoardHub: A Smart Boarding House Management Information System
A research project that addresses the need for a comprehensive solution in managing boarding houses by integrating a web app with Internet of Things.

ESP32 SmartHome Security
An Internet of Things project for home surveillance and security using ESP32 and ESP32-CAM with two servo motors for pan and tilt control. The user can monitor the video stream from anywhere with an internet connection.

Smart Doorbell System
An Internet of Things project for home surveillance and security using ESP32, ESP32-CAM, some sensors, and modules. This system offers real-time notification and remote monitoring using Blynk.

ESP32 Web Server with WebSockets
An IoT and web application development project for real-time control of components and visualization of sensor data using websockets.

Roll N' Shoot
A simple 3D shooter game written in C++ in Unreal Engine for a solo game development project.

Student Login and Registration System
A programming project written in C++ utilizing Windows Forms .NET with filesystem.

Darkness Sensor Project
An electrical circuits final project for automated home lighting using a photoresistor, potentiometer, NPN transistor, and a resistor.

Tic Tac Toe Console Game
A programming game development final project written in C language. The game can be played against a computer or with another player in a command line console.

Research

AIoT Security System For Boarding House: Development of Vision-based Security System Using Facial Recognition
A thesis about the development of an AIoT Security System for boarding houses using advanced facial recognition algorithms (MTCNN, Inception-ResNet-V1) to enhance the security of boarding houses near Mindanao State University - Iligan Institute of Technology.

My.BoardHub: Development of a Single-Board Computer-Based Security System Using Artificial Intelligence of Things for Boarding Houses
Developed an AIoT-based security system integrating facial recognition on Raspberry Pi for real-time boarding house surveillance, achieving over 94% recognition accuracy and near real-time performance.
Published in: Novel and Intelligent Digital Systems: Proceedings of the 5th International Conference (NiDS 2025), October 2025.
DOI: 10.1007/978-3-032-06634-3_6